Problem Set 1: Simple Random Samples and Statistics

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Exercise 1

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Let $(X_1, X_2, \cdots, X_n)$ be an s.r.s. (n) of a population X with density function $f(x)$.

Determine the sampling distributions and the corresponding density functions of the statistics

(1) $T_1(X_1, X_2, \cdots, X_n) = \max(X_1, X_2, \cdots, X_n) = X_{(n)}$,

(2) $T_2(X_1, X_2, \cdots, X_n) = \min(X_1, X_2, \cdots, X_n) = X_{(1)}$.

Exercise 2

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Let $(X_1, \cdots, X_n)$ be an s.r.s. (n) of a uniform distribution in $(\theta - 1/2, \theta + 1/2)$.

(1) Determine the sampling distributions and the corresponding density function of the minimum $X_{(1)}$ and of the maximum $X_{(n)}$.

(2) Determine, moreover, its expectations and deduce a non-zero statistic whose expectation is zero.

Exercise 3

A football pools die has the outcomes 1, 2, and X marked on it. To make inferences about the probabilities p_1, p_2, and p_X of the three possible outcomes, the die is rolled three times.

(1) Construct the sample space for a single roll of the die.

(2) Find the distribution of the statistic (T_1, T_2, T_X), where T_i is the frequency of the results equal to i.

(3) Determine the distribution of T_X, its expectation and its variance.

(4) Compute the covariance between T_1 and T_2.

Exercise 4

For a population with Poisson distribution of parameter $\lambda > 0$ an s.r.s. of size $n \geq 1$ is obtained.

Determine the sampling distribution of the sample mean.

Exercise 5

Let X be a r.v. that takes the values $1, 2, 3$ and 4 with respective probabilities $1/8, 2/8, 4/8$ and $1/8$. We extract an s.r.s. (2) of this random variable and we compute the average value of the values observed.

Compute the sample distribution as well as the distribution of the statistic $\overline{X} = (X_1 + X_2)/2$, the sample mean.

Exercise 6

Determine the sample distribution of $\overline{X}$ and S^2 corresponding to an s.r.s. of size $n \geq 1$ of the random variable

$$X = \begin{cases} 1 & \text{with probability } \theta, \\ 0 & \text{with probability } 1-\theta. \end{cases}$$

Exercise 7

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Let (X_1, X_2, X_3) be an s.r.s. (3) of a population with density function $f(x) = 2x$ if $0 < x < 1$ and 0 otherwise.

Compute the probability that the statistic $X_{(1)}$ exceeds the median of the distribution.

Exercise 8

Let $(X_1, \cdots, X_6)$ be an s.r.s. (6) of a characteristic X of a population $\mathcal{N}(\mu, \sigma^2 = 12)$.

Compute $P\left(2.3 < \mathcal{S}^2 < 22.2\right)$, where $\mathcal{S}^2$ is the unbiased sample variance.

Exercise 9

Let $(X_1, \cdots, X_{25})$ be an s.r.s. of a characteristic X of a population $\mathcal{N}(3, \sigma^2 = 100)$.

Compute $P\left(0 < \overline{X} < 6; 5.2 < \mathcal{S}^2 < 145.6\right)$.

Exercise 10

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Let $(X_1, \cdots, X_{25})$ and $(Y_1, \cdots, Y_{25})$ be independent s.r.s. of two normal characteristics $\mathcal{N}(0, \sigma = 4)$ and $\mathcal{N}(1, \sigma = 3)$, respectively. Compute $P\{\overline{X} > \overline{Y}\}$.

Exercise 11

Compute the endpoints of the symmetric interval about the mean that excludes 5% of the observations from a $\mathcal{N}(\mu, \sigma)$ distribution.